

# Semantic-Aware Joint Source-Channel Coding for Ultra-Low-Latency Mobile Edge Networks

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## ABSTRACT

Mobile edge networks require a trade-off between communication cost and task accuracy. This paper presents a transparent semantic joint source-channel coding prototype that compresses 64-dimensional synthetic task features into an 8-dimensional latent representation and transmits it over a Rayleigh fading channel. The prototype is evaluated against a matched raw-feature baseline at 0, 2, 5, and 10 dB SNR. At 2 dB, the semantic representation uses 12.5% of the raw feature dimension and achieves 90.83% nearest-centroid task accuracy, while the raw baseline achieves 100%. At 0 dB, semantic accuracy falls to 83.89%. These results show a measurable bandwidth-accuracy trade-off, not a universal advantage. The artifact is a reproducible baseline for a subsequent trained Deep JSCC evaluation with public image/task data.

## KEYWORDS

semantic communication, joint source-channel coding, Rayleigh fading, mobile edge computing, 6G

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## 1 INTRODUCTION

Conventional communication stacks transport bits without knowing whether the receiver needs exact reconstruction or only a task-relevant representation. Semantic communication addresses this mismatch, but a useful evaluation must compare semantic compression against a matched raw baseline under the same channel. This work provides such a prototype for mobile edge networking.

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## 2 SYSTEM MODEL

An input feature vector  $x \in \mathbb{R}^{64}$  is standardized and projected into a latent representation  $z \in \mathbb{R}^8$ . The Rayleigh channel is:

$$\hat{z} = h \odot z + n, \quad n \sim \mathcal{N}(0, \sigma_n^2 I), \quad (1)$$

where  $h$  is sampled from a Rayleigh distribution and noise power is selected for the target SNR. The decoder applies the inverse projection and a nearest-centroid task classifier.

## 3 METHODS

The prototype uses PCA as an explicit semantic compression baseline. PCA is not a neural Deep JSCC model; it provides an interpretable latent bottleneck for testing the network/channel protocol before expensive learned training. The raw baseline transmits all 64 standardized features through the same channel. The semantic baseline transmits 8 features, yielding a dimension ratio of  $8/64 = 0.125$ .

## 4 EXPERIMENTAL PROTOCOL

We generate 1,200 labeled synthetic vectors from four class centers with Gaussian variation. The data are split using stratification and seed 20260915. The train/test split contains 840 and 360 samples. Experiments use SNR values 0, 2, 5, and 10 dB. The metric is nearest-centroid classification accuracy after channel corruption. This is a methodological proxy, not a real image or clinical dataset.

## 5 RESULTS

Table 1: Seeded semantic JSCC prototype results.

| SNR   | Semantic accuracy | Raw accuracy | Dimension ratio |
|-------|-------------------|--------------|-----------------|
| 0 dB  | 83.89%            | 100.00%      | 12.5%           |
| 2 dB  | 90.83%            | 100.00%      | 12.5%           |
| 5 dB  | 96.39%            | 100.00%      | 12.5%           |
| 10 dB | 97.78%            | 100.00%      | 12.5%           |

The prototype demonstrates compression with graceful but measurable accuracy loss at low SNR. It does not reproduce the earlier unverified claim of 72% bandwidth reduction and 95.4% task accuracy; those claims require a trained model and public task dataset.

## 6 LIMITATIONS AND FUTURE WORK

The dataset is synthetic, PCA is not Deep JSCC, the classifier is simple, and latency is not measured on a mobile-edge hardware

platform. Future work will use public image/task data, a learned encoder/decoder, explicit power normalization, Rayleigh and blockage traces, repeated seeds, and latency/energy measurements.

## 7 CONCLUSION

This work establishes a reproducible semantic JSCC baseline for INCC 2026 and makes the compression–accuracy trade-off measurable. Its principal value is methodological: it prevents a semantic communication claim from being made without a matched channel

baseline. A trained model and public-data experiment are required for the final submission.

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